



RumorGAL: Graph-anchored LLM for domain-generalized rumor detection

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ABSTRACT

Rumor detection models must remain robust to breaking events that are unseen during training. Although large language models (LLMs) offer strong generalization potential, their text-centric nature makes them vulnerable to spurious textual patterns in noisy social media discussions and leaves them without explicit awareness of how rumors propagate through conversation threads. To address these limitations, we propose RumorGAL, a graph-anchored LLM framework for domain-generalized rumor detection. The framework first employs a graph neural network (GNN) to derive graph-anchored structural prefixes from the propagation graph, providing structural guidance for the LLM. It then identifies high-quality textual evidence from noisy discussions through an evidence density-guided mining strategy. Finally, a hierarchical reasoning mechanism organizes both the structural prefixes and textual evidence into a unified reasoning process, enabling the LLM to ground its predictions in both propagation patterns and informative content. Across the three transfer settings, RumorGAL achieves robust performance, improving average accuracy over the strongest baseline, zero-shot GPT-4.1, by 5.27 percentage points under the domain generalization setting. When compared with domain adaptation methods, it surpasses the best domain adaptation method in two out of three tasks, with improvements of 3.1 and 0.5 percentage points in F_1 score for the rumor class, respectively. These findings suggest that an *LLM-as-primary*, *GNN-as-auxiliary* design, which couples structure-aware guidance with evidence-grounded reasoning, is a promising approach to robust and interpretable rumor detection.

1. Introduction

During breaking events, social media platforms have become major channels for information sharing, but they also facilitate the rapid spread of rumors and misinformation before authoritative verification is available, posing serious risks to public health, politics, and economic activity (Dong et al., 2025; Harney, 2006; Petersen et al., 2023). This challenge became particularly evident during the COVID-19 pandemic, when unverified claims such as “chloroquine and ivermectin cure COVID-19”¹ and “drinking bleach eliminates the virus”² rapidly circulated through online discussions, intensifying public anxiety and disrupting timely public health responses (Frey et al., 2024; Gao et al., 2020; Islam et al., 2020). This situation requires rumor detection models to remain robust to unseen breaking events for which no target-domain samples are available during training, making domain-generalized rumor detection a practically important yet challenging task.

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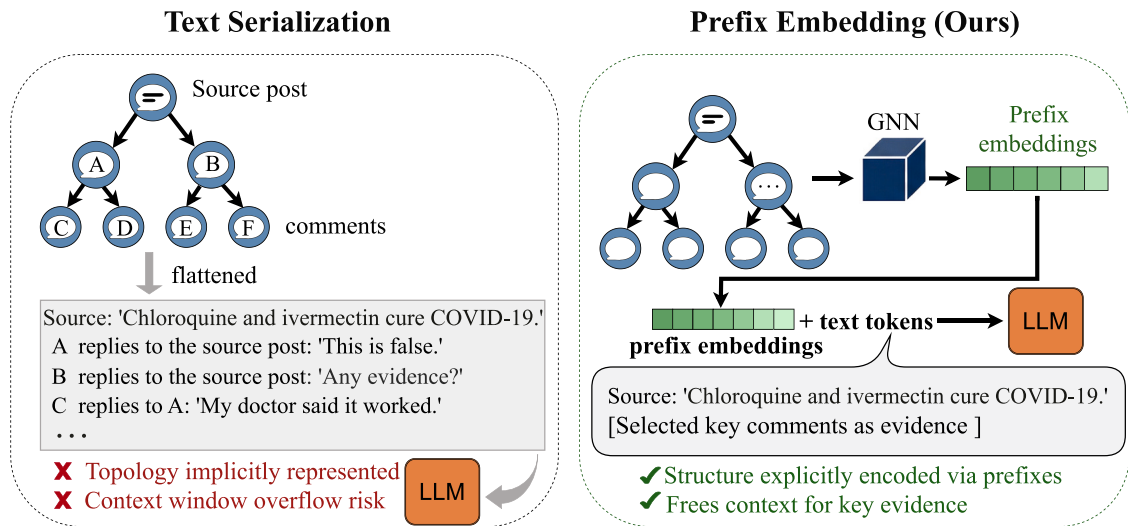


Fig. 1. Comparison of text serialization and prefix-based encoding for incorporating propagation structure into an LLM.

Early feature-based (Castillo et al., 2011; Kwon et al., 2013; Qazvinian et al., 2011; Vicario et al., 2019) and deep learning methods (Ma et al., 2016, 2019; Yu et al., 2017) rely primarily on textual content, linguistic cues, user attributes, or temporal patterns, but lack the ability to model propagation structure within discussion threads. This limits their capacity to capture rumor diffusion dynamics and generalize across unseen events. To address this, recent methods (Bian et al., 2020; Cui & Jia, 2024; Li et al., 2019; Qiao et al., 2024) employ graph neural networks (GNNs) to model conversation structures, thereby capturing reply relations and diffusion patterns beyond textual content. However, their generalization to unseen events remains limited, as these methods mainly rely on propagation patterns while making insufficient use of the broad semantic priors afforded by large-scale pretraining and the fine-grained evidential cues embedded in noisy user responses.

Domain adaptation methods (Liu et al., 2025; Ran & Jia, 2023; Shi et al., 2024) based on feature alignment or pseudo-labeling can effectively improve rumor detection across domains. However, they typically rely on target-domain information during training or deployment, which limits their applicability in practical scenarios where such information cannot be accessed beforehand. Recent large language models (LLMs) offer a promising alternative, benefiting from internalized world knowledge acquired during large-scale pretraining (Budnikov et al., 2025; Chang et al., 2024; Ouyang et al., 2022). Existing LLM-based rumor detection methods can be broadly divided into two categories. The first category uses LLMs as auxiliary modules to inject external knowledge, generate rationales, or provide supervision signals for downstream detectors (Chen et al., 2025; Hu et al., 2024; Wu et al., 2024). The second category places LLMs at the core of the detection pipeline as the main reasoning component (Cheung & Lam, 2023; Q. Liu et al., 2024; Zeng et al., 2026). Although the above methods have achieved encouraging results, most existing LLM-based rumor detection approaches remain predominantly text-centric, with limited incorporation of propagation structure. Consequently, they are more likely to rely on superficial patterns in linearized text, which are easily reinforced in social media environments characterized by fragmented interactions, emotional expression, and rapidly evolving discussions, thus weakening robustness in noisy conversational settings.

In summary, the existing literature reveals several critical limitations of current rumor detection methods under domain-generalized settings: (1) existing LLM-based approaches do not explicitly incorporate propagation structure when target-domain data are unavailable; (2) GNN-based methods can capture propagation patterns, but they make limited use of the broad semantic knowledge afforded by large-scale pretraining; (3) domain adaptation methods depend on access to target-domain data, which is often unavailable in the early stages of breaking events; and (4) current LLM-based methods remain largely text-centric, making them vulnerable to noisy surface cues. Motivated by these limitations, this study aims to develop a domain-generalized framework for breaking events that learns solely from source-domain data while achieving robust performance on unseen target events through the joint use of propagation structure and LLM-based reasoning.

To achieve this objective, we propose RumorGAL, a graph-anchored LLM framework for domain-generalized rumor detection. It adopts an *LLM-as-primary, GNN-as-auxiliary* design to jointly exploit semantic reasoning and propagation structure. As illustrated in Fig. 1, propagation structure can be incorporated into an LLM either by verbalizing a discussion tree as natural language or by encoding the propagation graph into compact prefix embeddings. However, textualization exposes structural information indirectly through serialized texts, which provides limited structural guidance for reasoning over long and complex social discussions. It also increases input length and reduces the space available for salient evidence under constrained context windows. We therefore adopt the latter strategy, encoding a propagation graph as compact prefix embeddings that are used together with selected textual evidence as complementary inputs to an LLM.

To realize this design, RumorGAL consists of three tightly coordinated components: structural anchoring (SA), evidence density-guided mining (EDM), and hierarchical reasoning chains (HRC). At the structural level, SA employs a graph neural network (GNN)

to encode the reply tree as a propagation graph, identify comment nodes that are structurally central to the source post, and project their graph representations into compact prefix embeddings. These embeddings are then prepended to the LLM input as lightweight structural cues. At the evidence level, EDM selects comments according to their semantic relevance to the source post and lexical informativeness, measured by cosine similarity and lexical entropy, respectively, thereby extracting fine-grained and informative evidential clues to support downstream reasoning. Building on these two components, HRC incorporates the structural prefix embeddings produced by SA together with the textual evidence selected by EDM into a unified LLM-centered reasoning framework. More specifically, through a Case Overview Prompt (COP), the model is provided with the source post, the selected evidence, and the structural prefixes as contextual support, while a Chain-of-Clue Prompt (CCP) further guides claim-level analysis and cross-evidence evaluation. In this way, RumorGAL enables the LLM to generate evidence-grounded predictions that rely primarily on high-quality textual evidence while exploiting propagation structure as auxiliary support, without requiring target-domain data during training. Experiments on three public transfer settings demonstrate that RumorGAL consistently outperforms representative baselines across multiple evaluation metrics.

The main contributions of this work are summarized as follows:

- We propose RumorGAL, a graph-anchored LLM framework for domain-generalized rumor detection that integrates high-quality textual evidence with auxiliary propagation-structure cues.
- We develop three coordinated mechanisms for graph-anchored evidence reasoning. SA maps source-post-centered graph representations into continuous structural prefixes for the LLM. EDM selects informative comments using event-normalized semantic relevance and lexical entropy. HRC guides the LLM to perform systematic claim verification and cross-evidence assessment over these structural and textual inputs.
- We conduct extensive experiments on three publicly available transfer settings. Results show that RumorGAL consistently outperforms representative baselines, achieving an average gain of 5.27 percentage points over zero-shot GPT-4.1 in the domain generalization setting and outperforming the strongest domain-adaptation baseline in two of the three transfer settings, without using any target-domain data during training.

The remainder of this paper is organized as follows. Section 2 reviews related work on rumor detection. Section 3 introduces the problem setup and explains the design rationale of the proposed framework. Section 4 presents the proposed framework in detail. Section 5 describes the datasets, baselines, and implementation details. Section 6 reports the main results and further analyses of the model's components, robustness, temporal generalization, efficiency, and cases. Section 7 discusses the findings and presents the theoretical implications, practical implications, and limitations of this study. Section 8 concludes the paper.

2. Related work

2.1. Deep learning-based rumor detection

Early rumor detection methods mainly relied on manually designed textual and user-related features, such as credibility cues, sentiment signals, and lexical patterns (Castillo et al., 2011; Kwon et al., 2013; Qazvinian et al., 2011; Vicario et al., 2019). However, such methods often required labor-intensive feature engineering and had limited ability to generalize across events. With the rise of deep learning, recurrent networks, convolutional architectures, and generative frameworks were introduced to learn richer contextual representations from textual sequences and temporal dynamics, leading to notable improvements in rumor and misinformation detection (Ma et al., 2016, 2019; Yu et al., 2017). However, these methods still lacked explicit modeling of propagation topology, making it difficult to capture the structural dynamics of rumor dissemination. Multimodal rumor detection has also received increasing attention. A recent survey summarizes multimodal fake-news detection from the perspective of cross-modal interactions (Li et al., 2025). Recent multimodal rumor detection studies improve evidence modeling through text-image evidence enhancement, cross-modal similarity modeling, cross-modal content correlation, and few-shot multimodal meta-learning (Huang, Ma, Rong, et al., 2025; Qiao et al., 2025; Ye et al., 2026).

Graph-based rumor detection complements textual semantics by modeling propagation structures and user interactions. Early structural studies focused on propagation topologies, using tree-structured recursive neural networks and bidirectional graph convolutional networks to capture top-down propagation and bottom-up dispersion (Bian et al., 2020; Ma et al., 2018). Building on this foundation, subsequent methods improved structural representations through context-specific path aggregation (Zhang et al., 2021), noise suppression in propagation graphs (Zhu et al., 2024, 2026), and adaptive spectral graph filtering (Yu et al., 2025). Related work has also examined how graph structures contribute to prediction explainability (Chin & Lee, 2026a, 2026b; Chin et al., 2024). To improve robustness under unreliable participation, uncertainty-aware diffusion modeling further captures the propagation of uncertainty through fragile topologies (He et al., 2026). Alongside topology-centric efforts, another line of research incorporates user interactions and temporal dynamics into diffusion analysis. Multiview models jointly capture the evolution of information propagation and user interaction graphs (Huang, Ma, Jin, et al., 2025), while behavior-oriented approaches distinguish the roles of different followers in the diffusion process (Wu et al., 2026). Stochastic hybrid frameworks further investigate intervention strategies for dynamic rumor control (Zhong et al., 2025). Recent studies have further incorporated user cognition by modeling multisource resonance and cognitive adaptation (W. He et al., 2025), and by formulating rumor and antirumor interactions as a cognitive game to characterize user state transitions (Mou et al., 2025). Together, these studies highlight the importance of structural dynamics and user behavior in rumor detection. However, propagation-based methods often exhibit limited generalization to unseen domains.

To address this limitation, one line of research focuses on domain adaptation for rumor detection. Lu et al. (2022) introduced a reinforcement-learning-based sequential training strategy to reduce the influence of noisy comments, while Lu et al. (2023) combined meta self-training with domain adversarial initialization to improve pseudo-label selection and source–target adaptation. Shi et al. (2024) transferred source knowledge to the target domain by disentangling emotional and semantic features, Z. He et al. (2025) developed a gradient coherence guided meta-learning framework to alleviate target-domain sample scarcity in emerging-topic rumor detection, and Ran et al. (2026) learned domain-invariant representations through cross-attention over source and target instances with matched labels. Although these methods have shown effectiveness in cross-domain settings, they rely on access to target-domain data for adaptation. By contrast, our work considers domain-generalized rumor detection, in which the model must generalize to unseen target domains without access to target-domain data.

2.2. LLM-based rumor detection

Recent advances in LLMs suggest that large-scale pretraining can provide broad semantic knowledge, strong reasoning capabilities, and more generalizable representations, making LLMs increasingly attractive for rumor detection (Chang et al., 2024; DeepSeek-AI, 2025; Ouyang et al., 2022). Existing LLM-based rumor detection methods can be broadly divided into two categories.

The first category uses LLMs as auxiliary tools to enhance specialized rumor detection models. The common goal of this line of work is to exploit the knowledge-generation and reasoning capabilities of LLMs to support downstream detectors. For example, Wu et al. (2024) used LLMs to generate style-camouflaged fake news to improve style-robust veracity prediction. Wang et al. (2024) and Chen et al. (2025) leveraged LLM reasoning to generate external factual knowledge for downstream models. Hu et al. (2024) showed that inadequate rationale integration can weaken detection performance and proposed the Adaptive Rationale Guidance Network (ARG), in which small language models (SLMs) exploit LLM-generated rationales. Similarly, Yin et al. (2026) used LLM-generated truthfulness and falseness reasons with contrastive learning and adaptive boosting for fake-news detection. Yang et al. (2023) used ChatGPT for knowledge enhancement to support semantic feature learning, while H. Liu et al. (2024) combined human expertise with LLM-generated logical atoms to improve neural-symbolic decision-making. These studies show that LLMs can provide useful external knowledge, rationales, and supervision signals. However, in this category, LLMs mainly serve as auxiliary components for knowledge enhancement or rationale generation, while the final prediction is still made by a separate downstream detector. This makes the role of LLMs indirect and may limit the extent to which their semantic and reasoning capabilities can be utilized.

The second category treats LLMs as core components of the rumor detection pipeline, aiming to use them directly for evidence understanding and decision-making. For example, Cheung and Lam (2023) fine-tuned a LLaMA-based (Touvron et al., 2023) detector with LoRA (Hu et al., 2022) and external factual knowledge. Pan et al. (2023) employed in-context learning to generate executable reasoning programs, improving both interpretability and data efficiency. Q. Liu et al. (2024) guided LLMs to reason over key clues in news and comments by organizing propagation information into a chain of propagation. Wan et al. (2024) incorporated LLMs into multiple stages of the misinformation detection pipeline, while Yang et al. (2024) used LLMs for joint stance detection and rumor verification with reinforcement-based data selection and tuning. These studies demonstrate the promise of LLM-centered rumor detection. However, most of them remain largely text-centric and do not explicitly integrate propagation structure into LLM-based reasoning.

2.3. Integrating GNNs and LLMs for text-attributed graphs

Recent studies on combining GNNs and LLMs for text-attributed graphs provide useful methodological insights for our method. Existing work can be broadly grouped into two categories. The first category investigates whether graph information can be translated into textual form and directly processed by LLMs. For example, Ye et al. (2024) represented multi-scale graph structure in natural language and instruction-tuned the LLM for graph tasks, while Zhao et al. (2024) converted graphs into graph-syntax trees and then linearized them into text sequences for graph reasoning. Further analysis by Huang et al. (2023) showed that structural information can improve LLM performance on node classification. However, such textualization-based methods may obscure explicit graph topology. The second category studies how to inject graph structure into LLMs through graph-to-token projection or graph-aware prompting. Tang et al. (2024) integrated graph structural knowledge into LLMs through text–graph grounding and dual-stage graph instruction tuning, thereby improving generalization across diverse graph datasets and tasks. Tian et al. (2024) proposed Graph Neural Prompting (GNP), a plug-and-play framework that enhances pretrained LLMs with grounded knowledge from knowledge graphs through graph neural prompting, improving reasoning performance without retraining the full language model. Qin et al. (2024) enhanced LLM-based learning on text-attributed graphs by introducing tailored disentangled GNN layers to inject structural information into frozen pretrained LLMs, thereby improving reasoning, prediction, and interpretability. This line of research provides important methodological guidance for incorporating structural information into LLMs.

Nevertheless, their application to rumor detection under domain generalization remains limited. Unlike general text-attributed graph tasks, this setting requires the model to jointly capture propagation-aware structural signals and perform nuanced semantic reasoning over source posts and user discussions. Building on this insight, our method introduces graph-anchored prefix representations to incorporate structural cues into the LLM reasoning process.

Bridging critical gaps in prior work. Existing approaches exhibit three interrelated limitations in domain-generalized rumor detection. Structure-oriented methods primarily use propagation graphs as predictive representations but provide limited semantic and evidential reasoning, whereas LLM-based approaches offer strong reasoning capabilities but remain largely text-centric. Cross-domain rumor detection methods commonly rely on target-domain data for adaptation, which may be unavailable for

emerging events. RumorGAL addresses these limitations through an *LLM-as-primary*, *GNN-as-auxiliary* design. Specifically, it encodes propagation structure as compact structural prefixes that condition evidence-grounded LLM reasoning rather than serving as the final prediction representation. RumorGAL is therefore distinguished by its use of propagation structure as an auxiliary reasoning prior within an LLM-centered detector under a strict domain-generalization setting that requires no access to target-domain data.

3. Problem setup and design rationale

Task formulation. A social media event is represented as $\mathcal{E} = (s, C, G)$, where s denotes the source post, $C = \{c_1, c_2, \dots, c_{|C|}\}$ denotes the associated comment set, and $G = (V, A)$ denotes the propagation graph. The node set is defined as $V = \{v_s, v_{c_1}, \dots, v_{c_{|C|}}\}$, where v_s corresponds to the source post and v_{c_i} corresponds to comment c_i . The adjacency matrix A encodes the reply relations in the propagation graph. Under the domain generalization setting considered in this study, only labeled events from the source domain are available during training. The source-domain dataset is defined as $\mathcal{D}_s = \{(\mathcal{E}_i, y_i)\}_{i=1}^{|\mathcal{D}_s|}$, where $y_i \in \{\text{Rumor}, \text{Non-Rumor}\}$. The objective is therefore to learn a prediction function $f_\theta(\mathcal{E})$ from $\mathcal{D}_s$ that generalizes to unseen target-domain events.

Design Motivation. Existing studies still leave several unresolved issues for domain-generalized rumor detection. In particular, three key questions remain insufficiently addressed.

1. *How can propagation structure contribute beyond standalone graph representation learning?* Propagation-oriented GNN methods effectively model structural diffusion patterns, but their decisions rely heavily on graph representations and lack deep semantic and evidential reasoning over source claims and noisy discussions.
2. *How can LLM-based rumor detection move beyond a text-centric paradigm?* LLM-based approaches benefit from broad knowledge and flexible reasoning, yet most still depend primarily on source content, comments, and prompt design, without explicitly incorporating transferable propagation structure into the reasoning process.
3. *How can robust rumor detection be achieved without target-domain data?* Domain adaptation methods can bridge source–target discrepancies, but their reliance on target-domain data limits applicability to breaking events, where such supervision is rarely available in time.

Together, these limitations reveal a fundamental trilemma: GNN-based methods make limited use of semantic reasoning, LLM-based methods often underutilize propagation structure, and domain adaptation methods require access to target-domain data that may be unavailable for emerging events. This motivates a unified framework that jointly exploits propagation-aware structural information and LLM-based semantic reasoning under the domain-generalization setting.

Research Objective. Motivated by the above limitations, this study aims to develop a domain-generalized rumor detection framework for breaking events that learns solely from source-domain data while achieving robust performance on unseen target events through the joint use of propagation structure and LLM-based reasoning. In practical terms, the proposed RumorGAL framework is intended for emerging-event rumor detection scenarios where target-domain data are difficult to obtain in a timely manner, while its parameter-efficient fine-tuning design further improves deployability.

4. Method

This section provides an overview of RumorGAL, with the overall architecture illustrated in Fig. 2. RumorGAL comprises three closely connected components: the structural anchoring module (SA), the evidence density-guided mining module (EDM), and the hierarchical reasoning chains module (HRC). SA first derives source-post-centered structural anchors from the propagation graph and converts them into structural prefix embeddings. Based on the source post and comments, EDM then identifies informative evidence through proxy-based comment selection, while HRC formulates hierarchical reasoning prompts to guide evidence-grounded analysis. Finally, the structural prefixes, mined evidence, and reasoning prompts are integrated into a unified input for an LLM, which produces the final prediction. Details of these components are presented in the following subsections.

4.1. Structural anchoring

The SA module encodes the propagation graph with a graph encoder, constructs a source-post-centered structural anchor sequence, and projects the anchors into the LLM token embedding space as structural prefix embeddings.

Graph encoding. For a social media event with source post s , comment set $C = \{c_1, \dots, c_{|C|}\}$, and propagation graph $G = (V, A)$, the initial node feature matrix is $\mathbf{X} = [\mathbf{x}_s; \mathbf{x}_{c_1}; \dots; \mathbf{x}_{c_{|C|}}] \in \mathbb{R}^{|V| \times d_x}$, where each node feature is initialized by the BERT (Devlin et al., 2019) representation of the corresponding post or comment, and d_x denotes the representation dimension. The structure-aware node representations are obtained as

$$\mathbf{H} = \text{GNN}(\mathbf{X}, \mathbf{A}) = [\mathbf{h}_s; \mathbf{h}_{c_1}; \dots; \mathbf{h}_{c_{|C|}}] \in \mathbb{R}^{|V| \times d_g}, \quad (1)$$

where $\mathbf{h}_s, \mathbf{h}_{c_i} \in \mathbb{R}^{d_g}$ denote the GNN output representations of the source post and comment c_i , respectively, and d_g denotes the graph representation dimension. In this way, BERT provides semantic node features, while the GNN contextualizes them with propagation topology.

Anchor construction. To construct source-post-centered structural anchors, we rank comment nodes according to their cosine similarity with the source-post node in the graph-encoded space and select the top- k nodes as structural anchors. Specifically, the

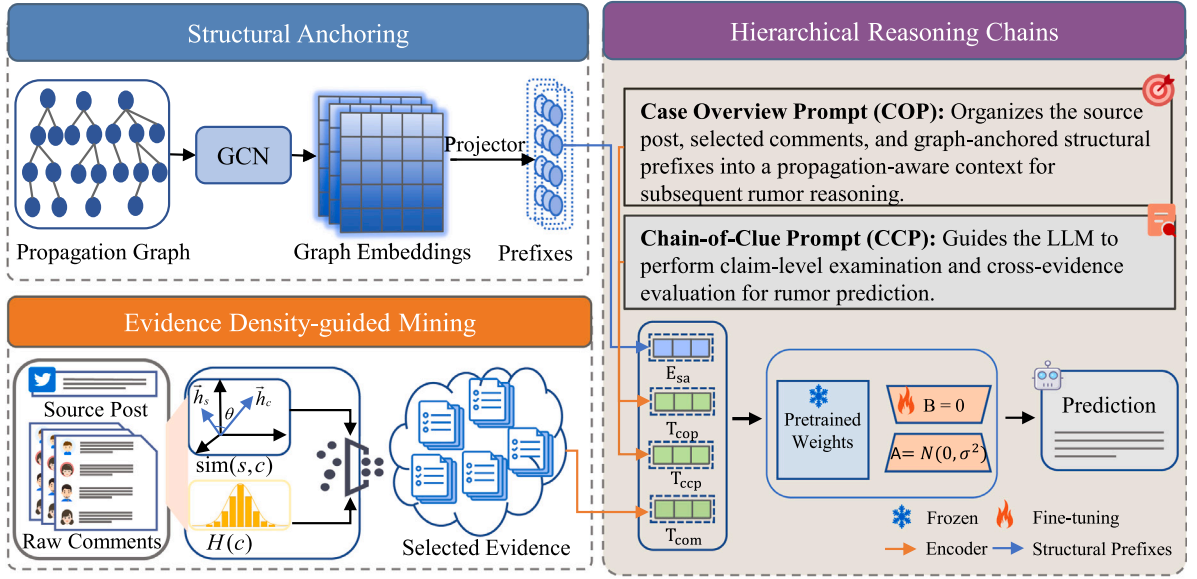


Fig. 2. Overview of the RumorGAL framework. The structural anchoring module derives graph-anchored structural prefixes from the propagation graph, while the evidence density-guided mining module selects informative comments from noisy discussions. These structural and textual signals are then integrated by the LLM through hierarchical reasoning chains for evidence-grounded rumor detection. Parameter-efficient fine-tuning enables efficient adaptation to the downstream task.

relevance score is computed as

$$S^{\text{str}}(c_i) = \frac{\mathbf{h}_s^\top \mathbf{h}_{c_i}}{\|\mathbf{h}_s\| \|\mathbf{h}_{c_i}\|}. \quad (2)$$

According to $S^{\text{str}}(c_i)$, the top- k comments are selected to form a source-post-centered structural anchor sequence:

$$\mathbf{Z} = [\mathbf{h}_s; \mathbf{h}_{c_{i_1}}; \dots; \mathbf{h}_{c_{i_k}}] \in \mathbb{R}^{(k+1) \times d_g}, \quad (3)$$

where k is a hyperparameter controlling the number of selected structural anchors, and $c_{i_1}, \dots, c_{i_k}$ are the top- k comments ranked by $S^{\text{str}}(c_i)$. The anchor sequence $\mathbf{Z}$ preserves source-post-centered interaction patterns while retaining fine-grained structural cues from the propagation graph.

Structural-prefix projection. To obtain structural prefixes compatible with the LLM, we use a learnable projection module to map the source-post-centered structural anchor sequence into the LLM token embedding space. Given $\mathbf{Z} \in \mathbb{R}^{(k+1) \times d_g}$, the projection is implemented as a two-layer MLP:

$$\mathbf{E}_{\text{sa}} = \sigma(\mathbf{Z}\mathbf{W}_1 + \mathbf{b}_1)\mathbf{W}_2 + \mathbf{b}_2, \quad (4)$$

where $\mathbf{W}_1 \in \mathbb{R}^{d_g \times d_m}$, $\mathbf{b}_1 \in \mathbb{R}^{d_m}$, $\mathbf{W}_2 \in \mathbb{R}^{d_m \times d_\ell}$, and $\mathbf{b}_2 \in \mathbb{R}^{d_\ell}$ are learnable parameters. Here, d_m is the hidden dimension of the projection layer, d_ℓ is the token embedding dimension of the LLM, and $\sigma(\cdot)$ denotes the GELU activation function.

Through a learnable projection module, the structural anchor sequence is mapped to $\mathbf{E}_{\text{sa}} \in \mathbb{R}^{(k+1) \times d_\ell}$, representing the structural anchors as prefix embeddings in the LLM token embedding space. By prepending these prefixes to textual token embeddings, RumorGAL forms a unified input embedding sequence in which self-attention jointly contextualizes propagation-aware structural cues and textual evidence.

4.2. Evidence density-guided mining

Feeding all comments into the LLM often introduces substantial redundancy and irrelevant content, which not only increases computational cost but also dilutes useful evidence. In rumor detection, useful evidence should be semantically related to the source post while also containing sufficiently informative textual content. Motivated by this observation, EDM selects a subset of comments that is more likely to provide informative evidence for rumor reasoning.

Since ground-truth labels are unavailable for newly emerging events during inference, the contribution of each comment to rumor detection cannot be directly evaluated. To address this limitation, EDM adopts a lightweight proxy-based comment selection strategy based on semantic relevance and lexical informativeness. Semantic relevance measures how closely a comment is related

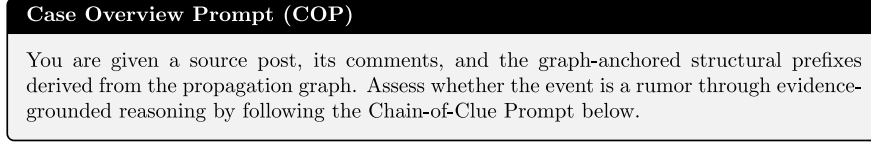


Fig. 3. Overview of the Case Overview Prompt (COP) structure.

to the source post, whereas lexical informativeness reflects whether the comment contains richer textual cues for rumor assessment rather than brief emotional reactions, repeated statements, or other low-information responses commonly observed in social media discussions.

The semantic relevance between the source post and comment c_j is measured by cosine similarity:

$$S^{\text{sem}}(c_j) = \frac{\mathbf{x}_s^\top \mathbf{x}_{c_j}}{\|\mathbf{x}_s\| \|\mathbf{x}_{c_j}\|}, \quad (5)$$

where $\mathbf{x}_s$ and $\mathbf{x}_{c_j}$ denote the text representations of the source post and comment c_j , respectively. Higher values of $S^{\text{sem}}(c_j)$ indicate stronger semantic consistency between the comment and the event context conveyed by the source post. To quantify the amount of information contained in each comment, lexical entropy is introduced as a word-level measure of lexical informativeness:

$$H(c_j) = - \sum_{w \in \mathcal{W}_{c_j}} p(w) \log_2 p(w), \quad (6)$$

where $\mathcal{W}_{c_j}$ denotes the set of words appearing in comment c_j , and $p(w)$ denotes the empirical probability of word w within that comment. Comments with greater lexical diversity generally yield higher entropy values, whereas short, repetitive, or template-like responses tend to exhibit lower entropy. However, lexical entropy alone is insufficient for identifying useful evidence, so it is combined with semantic relevance in Eq. (9).

Since semantic relevance and lexical entropy are defined on different scales, both quantities are normalized within each event over the corresponding comment set C :

$$\tilde{S}^{\text{sem}}(c_j) = \frac{S^{\text{sem}}(c_j) - \min_{c_i \in C} S^{\text{sem}}(c_i)}{\max_{c_i \in C} S^{\text{sem}}(c_i) - \min_{c_i \in C} S^{\text{sem}}(c_i) + \epsilon}, \quad (7)$$

$$\tilde{H}(c_j) = \frac{H(c_j) - \min_{c_i \in C} H(c_i)}{\max_{c_i \in C} H(c_i) - \min_{c_i \in C} H(c_i) + \epsilon}, \quad (8)$$

where the minimum and maximum are taken over the comment set of the current event, and ϵ is a small constant introduced for numerical stability, particularly when all comments share the same score. This event-wise min-max normalization mitigates scale inconsistency between the two indicators and enables their subsequent weighted aggregation.

Based on the normalized scores, the evidence density score of comment c_j is defined as

$$\text{Score}(c_j) = \alpha \tilde{S}^{\text{sem}}(c_j) + (1 - \alpha) \tilde{H}(c_j), \quad (9)$$

where $\alpha \in [0, 1]$ is a trade-off coefficient that balances semantic relevance and lexical informativeness. This weighted formulation prevents entropy from acting as an independent selection criterion. A comment with high lexical entropy but weak semantic relevance to the source post, such as an off-topic advertisement or word-stuffing response, will still receive a limited overall score. Conversely, comments are favored when they are both semantically aligned with the source claim and lexically informative.

After computing $\text{Score}(c_j)$ for all comments in C , the top- q comments are selected to construct the evidence sequence:

$$T_{\text{com}} = [s; c_{j_1}; \dots; c_{j_q}], \quad (10)$$

where $c_{j_1}, \dots, c_{j_q}$ denote the top- q comments ranked by $\text{Score}(c_j)$, and q is a hyperparameter controlling the number of retained evidence comments.

Through this process, EDM transforms a large and noisy comment set into a compact evidence sequence that retains comments with stronger event relevance and richer textual content. This design helps reduce the impact of redundant or off-topic comments, improves the utilization of the limited input budget of the LLM, and provides a more informative evidence basis for subsequent rumor reasoning.

4.3. Hierarchical reasoning chains

While the SA and EDM modules provide propagation-aware context and informative evidence, reliable rumor detection further benefits from structured reasoning guidance over the extracted information. To this end, we introduce HRC, a prompt-based reasoning scheme that organizes event-level context and evidence-level verification into a hierarchical reasoning process for rumor detection, as illustrated in Figs. 3 and 4.

Chain-of-Clue Prompt (CCP)

To support your assessment, analyze the event through the following reasoning procedure:

I. Claim-level examination

Factual plausibility: Does the claim conflict with established knowledge?

Logical coherence: Does the claim contain internal inconsistencies?

Evidential grounding: Does the claim provide verifiable evidence?

II. Cross-evidence evaluation

Counter-evidence: Do the comments provide explicit refutations?

External support: Do the comments cite authoritative sources?

Collective stance: Do the comments collectively challenge the core claim?

Answer with only one word: Yes or No.

Answer:

Fig. 4. Detailed reasoning steps in the Chain-of-Clue Prompt (CCP).

The reasoning process is guided by two prompts defined at different levels. The case overview prompt (COP), denoted by T_{cop} , provides event-level reasoning guidance by instructing the model to jointly consider the source post, the filtered evidence sequence, and the structural prefixes produced by the SA module. In this way, the model is encouraged to ground its judgment in the overall event context rather than in isolated evidence fragments.

The chain-of-clue prompt (CCP), denoted by T_{ccp} , further refines this guidance through a structured verification protocol. Specifically, it encourages the model to analyze the event through two dimensions: claim-level examination and cross-evidence evaluation. The former emphasizes factual plausibility, logical coherence, and evidential support, while the latter highlights counter-evidence, external verification signals, and the collective stance expressed in comments. This structured prompting strategy encourages the model to integrate multiple sources of evidence and to ground its judgment in explicit verification cues. Together, T_{cop} and T_{ccp} form a global-to-local prompting scaffold that supports structured and evidence-grounded rumor detection.

4.4. Evidence-conditioned decision

In the final stage, the structural prefixes, reasoning prompts, and filtered evidence are integrated into an evidence-conditioned input for the LLM. For the i th event instance, the case overview prompt T_{cop} , the chain-of-clue prompt T_{ccp} , and the filtered evidence sequence T_{com} are first organized into a unified textual sequence:

$$T_i = [T_{\text{cop}}; T_{\text{ccp}}; T_{\text{com}}]. \quad (11)$$

The sequence T_i is then tokenized into token IDs and mapped by the embedding layer of the LLM to the token embedding sequence $\text{Embed}(T_i)$. The resulting textual embeddings are concatenated with the structural prefix embeddings $\mathbf{E}_{\text{sa}}$ along the sequence dimension to form the final input embedding sequence:

$$\mathbf{E}_i = [\mathbf{E}_{\text{sa}} \parallel \text{Embed}(T_i)], \quad (12)$$

where $\mathbf{E}_{\text{sa}} \in \mathbb{R}^{(k+1) \times d_e}$ denotes the structural prefix embeddings derived from the propagation graph, and $\text{Embed}(T_i)$ denotes the token embedding sequence corresponding to T_i .

To make explicit how the structural prefixes are utilized by the LLM, let $\mathbf{E}_i^{(0)} = \mathbf{E}_i$ denote the input sequence to the transformer. At the l th layer, the query, key, and value matrices are computed as

$$\mathbf{Q}_i^{(l)} = \mathbf{E}_i^{(l-1)} \mathbf{W}_Q^{(l)}, \quad \mathbf{K}_i^{(l)} = \mathbf{E}_i^{(l-1)} \mathbf{W}_K^{(l)}, \quad \mathbf{V}_i^{(l)} = \mathbf{E}_i^{(l-1)} \mathbf{W}_V^{(l)}, \quad (13)$$

and the self-attention output is given by

$$\text{Attn}(\mathbf{E}_i^{(l-1)}) = \text{softmax}\left(\frac{\mathbf{Q}_i^{(l)} (\mathbf{K}_i^{(l)})^\top}{\sqrt{d_h}}\right) \mathbf{V}_i^{(l)}, \quad (14)$$

where d_h denotes the attention head dimension. Since $\mathbf{E}_i$ contains both the structural prefix embeddings and the textual embeddings, the textual tokens can attend to the structural prefixes during self-attention, thereby incorporating source-post-centered structural signals into the subsequent reasoning process.

Table 1
Statistics of the cross-domain datasets.

	Weibo ^a	Weibo_COVID19 ^a	Twitter ^a	Twitter_COVID19 ^a	Terrorist ^b	Gossip ^b
# Threads	4649	399	1154	400	5940	485
# Nodes	1 956 449	26 687	60 409	406 185	83 860	15 225
# Non-rumors	2336	146	579	148	3907	116
# Rumors	2313	253	575	252	2033	369
# Avg. nodes	420	67	52	1015	15	32
Domain type	Open	Single	Open	Single	Single	Single

^a Dataset repository for Weibo, Weibo_COVID19, Twitter, and Twitter_COVID19.

^b Dataset repository for the Terrorist and Gossip.

Given the input embedding sequence $\mathbf{E}_i$, the LLM produces contextualized hidden representations:

$$\mathbf{U}_i = \text{LLM}(\mathbf{E}_i), \quad (15)$$

where $\mathbf{U}_i \in \mathbb{R}^{L_i \times d_e}$, and L_i denotes the input sequence length. Following the prompt design, the final decision is read from a designated answer position in the output sequence. Let $\mathbf{u}_i^{\text{ans}} \in \mathbb{R}^{d_e}$ denote the hidden representation in $\mathbf{U}_i$ at this position, and let $\mathbf{o}_i \in \mathbb{R}^{|\mathcal{V}|}$ denote the corresponding vocabulary logit vector produced by the LM head:

$$\mathbf{o}_i = \text{LMHead}(\mathbf{u}_i^{\text{ans}}), \quad (16)$$

where $\mathcal{V}$ is the vocabulary of the LLM.

For implementation, the two labels are mapped to binary indices, denoted by $\mathcal{Y} = \{0, 1\}$, where 1 denotes Rumor and 0 denotes Non-Rumor. Under the current prompt formulation, the verbalizer token “Yes” corresponds to Rumor, whereas “No” corresponds to Non-Rumor. A verbalizer function $v(\cdot)$ is then introduced to project the binary labels into the vocabulary space:

$$v(1) = \text{“Yes”}, \quad v(0) = \text{“No”}.$$

Accordingly, the label-specific logits are obtained by selecting the logits of the corresponding verbalizer tokens:

$$\mathcal{E}_i = [\mathbf{o}_i[v(0)], \mathbf{o}_i[v(1)]], \quad (17)$$

where $\mathbf{o}_i[v(0)]$ and $\mathbf{o}_i[v(1)]$ denote the logits assigned to the tokens “No” and “Yes”, respectively. The corresponding label distribution is then computed as

$$\mathbf{z}_i = \text{softmax}(\mathcal{E}_i), \quad (18)$$

where $\mathbf{z}_i = [z_{i,0}, z_{i,1}]$ gives the predicted probabilities of Non-Rumor and Rumor, respectively. The final prediction is given by

$$\hat{y}_i = \arg \max_{y \in \mathcal{Y}} z_{i,y}. \quad (19)$$

For efficient adaptation, Low-Rank Adaptation (LoRA) is employed during fine-tuning while the backbone parameters remain frozen. The model is optimized using the cross-entropy loss over the verbalizer-induced label distribution:

$$\mathcal{L}(\theta) = -\frac{1}{N} \sum_{i=1}^N \sum_{y \in \mathcal{Y}} y_{i,y} \log z_{i,y}, \quad (20)$$

where N denotes the number of training instances, $y_{i,y}$ is the one-hot indicator of whether instance i belongs to class y , and θ denotes the trainable parameters in the current fine-tuning stage.

5. Experimental setup

5.1. Datasets

We evaluate RumorGAL on three dataset groups spanning diverse topics and event types. For each source–target dataset pair, the first dataset serves as the source domain and the second serves as the target domain. Under the domain generalization setting considered in this study, target-domain samples are used solely for evaluation, and neither the target-domain samples nor their labels are available during training. All experiments are formulated as binary classification tasks with two labels: Non-Rumor (NR) and Rumor. Detailed dataset statistics are reported in Table 1.

- **Weibo → Weibo_COVID19** (Lin et al., 2022; Ma et al., 2016): We follow the Weibo and Weibo_COVID19 versions used by (Lin et al., 2022). In Weibo, non-rumor threads whose source-post content becomes empty after removing repost placeholders, mentions, and emoji-only expressions are excluded. In Weibo_COVID19, instances without a valid propagation thread containing at least two nodes are excluded during propagation-graph construction.

- **Twitter** → **Twitter_COVID19** (Lin et al., 2022; Ma et al., 2017): We adopt the Twitter and Twitter_COVID19 versions used by (Lin et al., 2022). Twitter covers open-domain events, whereas Twitter_COVID19 focuses on COVID-19-related events.
- **Terrorist** → **Gossip** (Ran & Jia, 2023; Zubiaga et al., 2016): Following Ran and Jia (Ran & Jia, 2023), we use two single-domain subsets derived from PHEME. Terrorist covers events such as *germanwings*, whereas Gossip focuses on speculative social events such as *putinmissing*.

5.2. Baselines

To validate the effectiveness of RumorGAL, we compare it with a comprehensive set of baselines:

Sequence modeling methods

- **LSTM** (Ma et al., 2016): A long short-term memory network that models the temporal dynamics between a source post and its comments.
- **CNN** (Yu et al., 2017): A convolutional neural network that extracts local textual features to construct event-level representations.
- **Rumor-GAN** (Ma et al., 2019): A generative adversarial framework in which the generator perturbs conversation threads to learn more robust representations.
- **C3N-text** (Qiao et al., 2025): A text-only adaptation of C3N using the Chinese-CLIP text encoder and a lightweight classifier.

Propagation-based methods

- **TD-RvNN** (Ma et al., 2018): A top-down recursive neural network that captures information propagation dependencies along a tree structure.
- **BU-RvNN** (Ma et al., 2018): A bottom-up recursive neural network that aggregates information from leaf nodes to the root.
- **Bi-GCN** (Bian et al., 2020): A bidirectional graph convolutional network that models both top-down and bottom-up rumor propagation patterns.
- **PLAN** (Khoo et al., 2020): A transformer-based post-level attention network that integrates propagation topology with semantic representations.
- **PPA-WAE** (Zhang et al., 2021): A framework that jointly models propagation patterns and textual semantics by combining path aggregation with a Wasserstein autoencoder.
- **PSGT** (Zhu et al., 2024): A propagation-aware graph transformer that introduces structural positional encodings to capture both local and long-range dependencies.
- **KPG** (Zhang et al., 2025): A reinforcement learning-based key propagation graph generator for rumor detection.

LLM-based methods

- **LeRuD** (Q. Liu et al., 2024): An LLM-based framework that uses propagation-chain prompting to identify key clues in news and comments.
- **ARG** (Hu et al., 2024): A rationale-guided method in which LLM-generated rationales are used to assist SLM-based rumor detection.
- **SheepDog** (Wu et al., 2024): An LLM-enhanced framework that generates adversarial training samples and pseudo-labels for veracity assessment.
- **SePro-U** (Zeng et al., 2025): An LLM-based rumor detection baseline that leverages refined social context for final rumor verification.
- **LFND** (Yin et al., 2026): An LLM-assisted detection baseline that incorporates LLM-generated rationales and contrastive learning into fake news classification.
- **GPT-4.1-ZS** (OpenAI, 2025): A representative zero-shot reasoning baseline representing the performance of a high-capacity closed-source model under prompt-based inference.
- **DS-Text-LoRA** (DeepSeek-AI, 2025): A text-only LoRA baseline that uses the same DeepSeek-R1-7B backbone as RumorGAL and is fine-tuned with the same source-domain supervision, but without SA, EDM, or HRC.

Domain adaptation methods

- **UCD-RD** (Ran & Jia, 2023): An unsupervised cross-domain rumor detection framework that combines contrastive learning with a cross-attention mechanism.
- **RPL** (Lin et al., 2023): A response-aware prompt learning framework for zero-shot cross-lingual and cross-domain rumor detection.
- **CDTT** (Ran et al., 2026): A domain adaptation method that aligns latent stance features across domains through stance-based contrastive learning.

5.3. Implementation details

We use DeepSeek-R1-7B (DeepSeek-AI, 2025) as the LLM backbone and a 2-layer GCN as the propagation encoder. The LLM backbone is fine-tuned via Low-Rank Adaptation (LoRA) (Hu et al., 2022) with rank $r = 8$, while all other LLM parameters remain

Table 2

Performance comparison with baselines under the domain generalization setting.

Method	Weibo → Weibo_COVID19			Twitter → Twitter_COVID19			Terrorist → Gossip		
	Acc.	NR.(F ₁)	R.(F ₁)	Acc.	NR.(F ₁)	R.(F ₁)	Acc.	NR.(F ₁)	R.(F ₁)
LSTM (Ma et al., 2016)	0.4157	0.4025	0.4279	0.4123	0.3396	0.4258	0.3308	0.3456	0.3031
CNN (Yu et al., 2017)	0.4209	0.3821	0.4383	0.4058	0.2847	0.4495	0.3254	0.3562	0.2968
Rumor-GAN (Ma et al., 2019)	0.4323	0.3961	0.4583	0.4196	0.3568	0.4312	0.3254	0.3548	0.2937
C3N-text (Qiao et al., 2025)	0.4744	0.3763	0.4269	0.4875	0.3444	0.4244	0.3680	0.3464	0.3881
TD-RvNN (Ma et al., 2018)	0.4790	0.4366	0.5478	0.4355	0.4009	0.4578	0.3359	0.3247	0.2816
BU-RvNN (Ma et al., 2018)	0.4518	0.3876	0.5053	0.4133	0.3858	0.4225	0.3268	0.3012	0.2156
Bi-GCN (Bian et al., 2020)	0.6123	0.4408	0.6811	0.5167	0.3172	0.4637	0.3484	0.3104	0.2741
PLAN (Khoo et al., 2020)	0.3844	0.4606	0.2833	0.4550	0.4760	0.4323	0.3079	0.4094	0.1640
PPA-WAE (Zhang et al., 2021)	0.5756	0.3533	0.6526	0.4679	0.4328	0.4846	0.3556	0.3833	0.2579
PSGT (Zhu et al., 2024)	0.6733	0.6521	0.6852	0.6153	0.4855	0.6837	0.5234	0.4975	0.4502
KPG (Zhang et al., 2025)	0.6426	0.6628	0.6204	0.6286	0.4842	0.6596	0.5258	0.4576	0.3907
ARG (Hu et al., 2024)	0.5523	0.3475	0.6593	0.5411	0.3754	0.6391	0.4928	0.2765	0.6095
LeRuD (Q. Liu et al., 2024)	0.5742	0.3028	0.6935	0.5885	0.2264	0.7211	0.5386	0.3875	0.5973
SheepDog (Wu et al., 2024)	0.6679	0.5153	0.7859	0.5757	0.3896	0.6333	0.7227	0.6817	0.7465
SePro-U (Zeng et al., 2025)	0.6547	0.5960	0.7641	0.6065	0.4952	0.6776	0.6361	0.6320	0.6402
LFND (Yin et al., 2026)	0.7495	0.7037	0.7885	0.5850	0.5046	0.6463	0.6340	0.6072	0.5400
GPT-4.1-ZS (OpenAI, 2025)	0.7236	0.6224	0.8063	0.6521	0.5142	0.7405	0.7435	0.6952	0.7846
DS-Text-LoRA (DeepSeek-AI, 2025)	0.7586	0.7436	0.7745	0.6325	0.5227	0.7203	0.7214	0.6566	0.7358
RumorGAL	0.8328	0.8015	0.8568	0.6822	0.5461	0.7835	0.7624	0.7036	0.8263

Table 3

Performance comparison with domain adaptation methods.

Method	Weibo → Weibo_COVID19			Twitter → Twitter_COVID19			Terrorist → Gossip		
	Acc.	NR.(F ₁)	R.(F ₁)	Acc.	NR.(F ₁)	R.(F ₁)	Acc.	NR.(F ₁)	R.(F ₁)
UCD-RD	0.7345	0.6129	0.8254	0.6650	0.4147	0.7649	0.8488	0.8304	0.8635
RPL	0.7408	0.6358	0.7976	0.6324	0.5342	0.7008	0.8308	0.8156	0.8422
CDTT	0.7494	0.6599	0.8016	0.6723	0.4688	0.7785	0.8698	0.8195	0.8827
RumorGAL	0.8328	0.8015	0.8568	0.6822	0.5461	0.7835	0.7624	0.7036	0.8263

Table 4

Statistical comparison analysis of accuracy between RumorGAL and GPT-4.1-ZS.

Transfer setting	Metric	Method	Mean ± Std	Significance testing	
				p-value	95% CI
Weibo → Weibo_COVID19	Acc.	RumorGAL	0.8328 ± 0.0065	< 0.001***	[0.0724, 0.1511]
		GPT-4.1-ZS	0.7236		
Twitter → Twitter_COVID19	Acc.	RumorGAL	0.6822 ± 0.0032	< 0.01**	[0.0231, 0.0372]
		GPT-4.1-ZS	0.6521		
Terrorist → Gossip	Acc.	RumorGAL	0.7624 ± 0.0027	< 0.001***	[0.0153, 0.0236]
		GPT-4.1-ZS	0.7435		

Note: Mean ± Std is reported across three random seeds for trainable models, while fixed zero-shot baselines are reported as single scores. The 95% CIs and p-values are computed using paired bootstrap resampling over the same target-domain test instances. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

frozen. Our trainable models were optimized for 10 epochs with a per-device batch size of 2 and gradient accumulation steps of 8 using the AdamW optimizer (Loshchilov & Hutter, 2019). We performed grid search over learning rates in $\{1 \times 10^{-5}, 5 \times 10^{-5}, 1 \times 10^{-4}, 5 \times 10^{-4}, 1 \times 10^{-3}, 5 \times 10^{-3}\}$ on the source-domain validation split. The maximum input length is set to 1024 tokens. Results of trainable models are averaged over three random seeds.

6. Results and analysis

6.1. Overall performance

We categorize the baseline methods into four groups: sequence modeling methods, propagation-based methods, LLM-based methods, and domain adaptation methods. Tables 2 and 3 summarize the overall performance comparison, where the best results in each table are highlighted in **bold**. Overall, RumorGAL achieves the best performance among all non-adaptation baselines across the three transfer settings and remains competitive against domain adaptation methods. These results demonstrate the effectiveness of combining propagation-aware structural anchoring with evidence-guided LLM reasoning for domain-generalized rumor detection. We further report a statistical reliability analysis relative to GPT-4.1-ZS across all three transfer settings in Table 4. The confidence

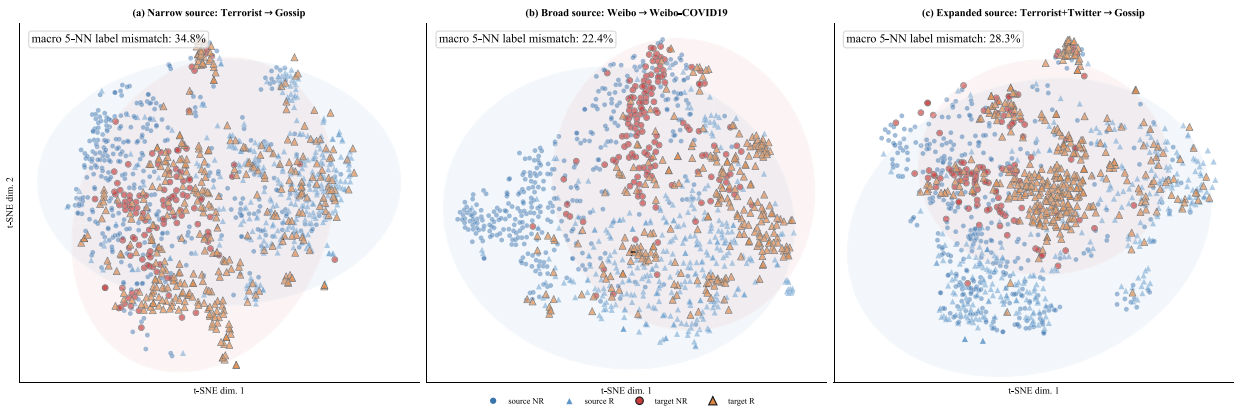


Fig. 5. t-SNE projection of thread-level structural-prefix representations.

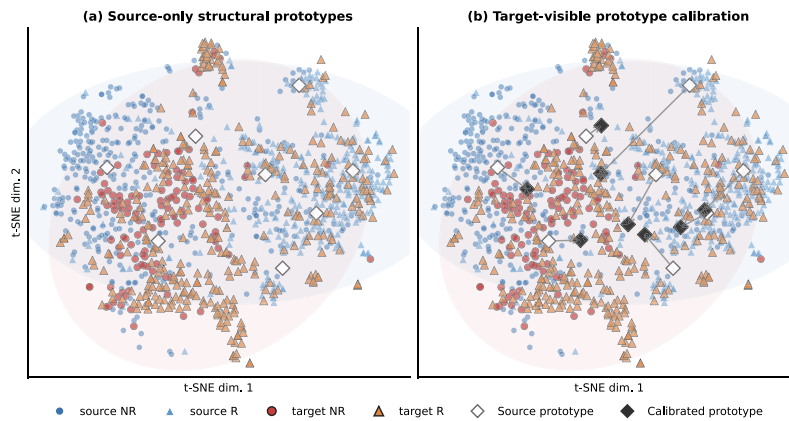


Fig. 6. Test-time target pseudo-prototype adjustment under Terrorist→Gossip transfer.

intervals and p -values are estimated by paired bootstrap resampling over target-domain test instances. The confidence intervals of the accuracy differences lie entirely above zero, indicating that the observed gains over GPT-4.1-ZS are statistically reliable.

The results reveal a broad trend among non-adaptation baselines, highlighting the complementary roles of structural modeling and semantic reasoning. The sequence modeling baselines generally underperform the other categories, suggesting that methods relying primarily on sequential textual features have limited capacity to capture robust and transferable cues. Propagation-based methods, by contrast, generally achieve stronger results, suggesting that explicit modeling of propagation structure can provide supplementary relational evidence beyond textual semantics. LLM-based baselines, such as GPT-4.1-ZS and LFND, further improve semantic generalization in several cases, but they do not explicitly model graph-level diffusion patterns or perform evidence-oriented comment selection. The improvement over DS-Text-LoRA further suggests that propagation-aware structural cues and evidence-guided reasoning are useful beyond text-only source-domain fine-tuning. Compared with the above baselines, RumorGAL introduces structural prefixes and hierarchical reasoning prompts to organize filtered evidence, enabling more evidence-grounded predictions. This advantage is particularly evident on Weibo→Weibo_COVID19 and Twitter→Twitter_COVID19, where RumorGAL achieves the best overall performance.

RumorGAL also remains competitive against domain adaptation methods, despite operating under the domain generalization setting. Compared with the domain adaptation methods reported in Table 3, RumorGAL outperforms the strongest domain-adaptation baseline on two of the three transfer settings, without using any target-domain samples during training. The exception is Terrorist→Gossip, where RumorGAL underperforms the domain adaptation baselines. This result suggests that extremely large semantic and social-context gaps may weaken structural transfer, especially when the source and target domains are both narrow single-event domains. It is also worth noting that domain adaptation methods exploit target-domain samples for distribution alignment, whereas RumorGAL operates without access to target-domain data.

Boundary analysis. We further analyze the challenging Terrorist→Gossip transfer case to characterize the limits of RumorGAL's generalization. As shown in Fig. 5, Terrorist→Gossip exhibits a higher macro 5-NN label-mismatch score than Weibo→Weibo_COVID19 and the combined Terrorist and Twitter source domain setting, indicating that many Gossip samples are close to Terrorist-domain neighborhoods with inconsistent labels. This suggests that limited source diversity constrains the transferability of structural patterns to unseen distributions, consistent with findings in domain generalization studies (Gulrajani &

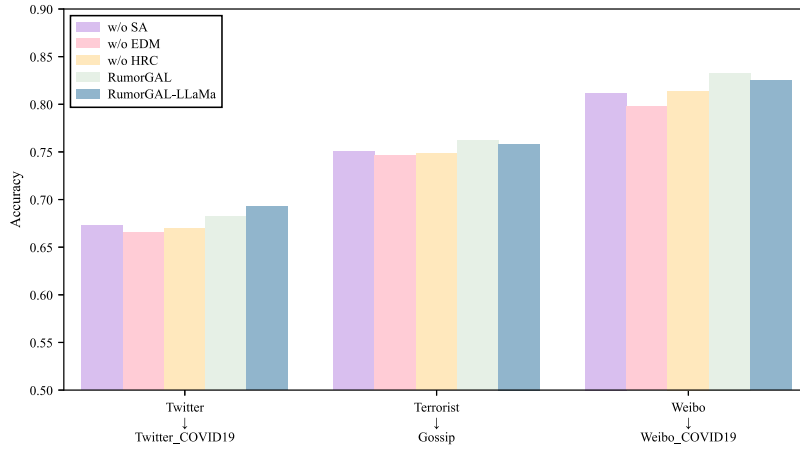


Fig. 7. Ablation results in terms of accuracy.

Table 5

Comparison of structural anchor selection strategies in terms of accuracy and rumor-class F_1 .

Strategy	Weibo→Weibo_COVID19		Twitter→Twitter_COVID19	
	Acc.	R.(F_1)	Acc.	R.(F_1)
Full graph mean pooling	0.7616	0.7860	0.6185	0.6630
Degree top- k	0.8100	0.8239	0.6686	0.7603
PageRank top- k	0.8002	0.8340	0.6337	0.7531
Cosine top- k (ours)	0.8328	0.8568	0.6822	0.7835

Lopez-Paz, 2021; Wei et al., 2022). We further perform a target-visible prototype calibration diagnostic in RumorGAL's structural-prefix space, as shown in Fig. 6. Following the target-aware calibration idea of CDTT, this diagnostic uses unlabeled Gossip samples to estimate target-visible prototypes and reveals how target-side information can alleviate structural mismatch.

6.2. Component and mechanism analysis

6.2.1. Ablation analysis

To evaluate the contribution of each component, we construct ablation variants by removing one module at a time while keeping the remaining RumorGAL pipeline unchanged. Specifically, **w/o SA** removes the source-post-centered structural prefix, **w/o EDM** replaces evidence-density-guided mining with chronologically selected comments, and **w/o HRC** removes the hierarchical rumor-checking instruction. We also replace the default LLM backbone with *LLaMA-3.1-8B* (Meta, 2024) to examine backbone dependence.

As shown in Fig. 7, removing any core component leads to consistent accuracy drops across the three transfer settings. The drop caused by **w/o SA** shows that structural prefixes provide useful propagation cues beyond textual evidence alone. The degradation of **w/o EDM** indicates that selecting compact evidence comments is important for reducing noisy and redundant discussions before LLM reasoning. The decline of **w/o HRC** further confirms the value of organizing rumor judgment through source-claim verification and comment-evidence integration. For the backbone replacement, *LLaMA-3.1-8B* achieves comparable performance to the default backbone, with improvements in some settings and slight decreases in others. This indicates that RumorGAL does not rely on a specific LLM backbone, while the final performance is still affected by backbone capacity and instruction-following ability.

6.2.2. Structural anchor selection analysis

We analyze structural anchor selection from two complementary perspectives by examining how different construction strategies affect domain generalization performance and where the selected anchors appear in the propagation tree. Table 5 compares four strategies, including full-graph mean pooling, degree top- k , PageRank top- k , and cosine top- k . Full-graph mean pooling yields lower performance than the top- k selection strategies, suggesting that averaging all comment-node representations into a single graph prefix may dilute discriminative propagation cues. Degree-based and PageRank-based selection improve over mean pooling but remain below cosine top- k in most metrics, indicating that global centrality alone is not sufficient for selecting effective structural anchors. To examine the tree positions of cosine-selected anchors, we use descendant subtree size as a simple local influence proxy, where a larger subtree indicates that a comment receives more downstream replies, following influence-spreading studies (Kitsak et al., 2010; Zhang et al., 2016). Fig. 8 shows that cosine-selected anchors are more likely to appear at locally influential discussion positions that receive follow-up replies. These results suggest that source-post-centered structural relevance provides a more effective criterion for constructing compact structural anchors.

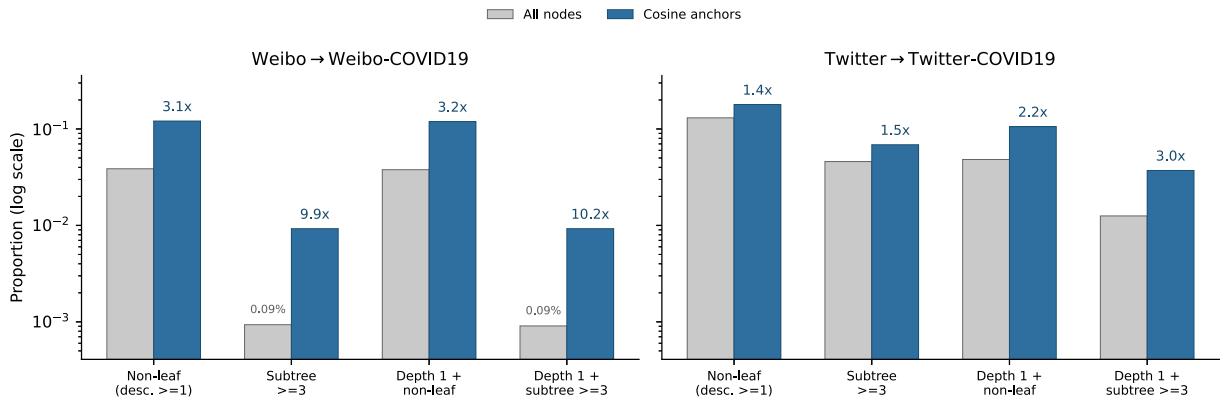


Fig. 8. Local influence of cosine-selected anchors. The figure compares cosine-selected anchors with all comment nodes using four reply-based indicators, including non-leaf comments, comments with at least three descendants, depth-1 non-leaf comments, and depth-1 comments with at least three descendants. Values above the blue bars denote the ratio between the selected-anchor proportion and the all-comment-node proportion.

Table 6
Comparison of graph backbones for structural anchoring.

Backbone	Params (M)	Weibo→Weibo_COVID19		Twitter→Twitter_COVID19	
		Acc.	R.(F ₁)	Acc.	R.(F ₁)
GCN (Kipf & Welling, 2017)	0.98	0.8328	0.8568	0.6822	0.7835
SGFormer (Wu et al., 2023)	1.64	0.8297	0.8606	0.6761	0.7797
BiGCN (Bian et al., 2020)	2.10	0.8168	0.8443	0.6673	0.7754
RAGCL (Cui & Jia, 2024)	4.06	0.8248	0.8506	0.6913	0.7928

6.2.3. Graph backbone analysis

To examine whether SA benefits from a more advanced graph encoder, we replace the default GCN with SGFormer (Wu et al., 2023), BiGCN (Bian et al., 2020), and RAGCL (Cui & Jia, 2024). The remaining RumorGAL pipeline is kept unchanged, including EDM, HRC, the LLM backbone, and the training protocol. The parameter count in Table 6 includes the structural graph encoder and excludes the shared LLM and LoRA modules. Replacing GCN with larger or more specialized graph encoders does not lead to consistent gains across transfer settings. GCN achieves the best accuracy on Weibo→Weibo_COVID19 with the smallest structural encoder size. The improvements from SGFormer and RAGCL are limited to specific metrics or datasets and require higher parameter costs. This result suggests that the main role of SA is not to perform rumor detection with a heavy graph classifier, but to extract structural anchors centered on the source post and provide compact propagation cues for the LLM. Therefore, a lightweight GCN provides a favorable balance among performance, efficiency, and architectural simplicity in our framework centered on the LLM.

6.2.4. Evidence-selection audit of EDM

In social-media discussions, noisy comments such as emoji-heavy responses, repeated-character strings, or promotion-like posts may also exhibit high lexical entropy. To examine whether this issue affects EDM selection, we analyze EDM at both the distribution and case levels. Fig. 9 visualizes candidate and EDM-selected comments in a two-dimensional plane defined by semantic relevance and lexical entropy. Selected comments are concentrated in the high-relevance/high-entropy region, with proportions ranging from 68.6% to 95.9% across datasets. By contrast, the high-entropy/low-relevance risk region accounts for at most 1.04% of selected comments. Table 7 further reports EDM decisions under three common noisy comment patterns, including emoji and character stuffing, repetitive text, and promotion-like comments. Across these patterns, noisy comments may obtain nontrivial entropy scores, yet EDM excludes them when they are only weakly related to the source event. These results indicate that lexical entropy serves as an auxiliary informativeness signal rather than a standalone criterion for evidence selection.

6.2.5. Effects of comment organization strategies

We further compare EDM with two comment organization strategies: chronological ordering and deep-shallow ordering. Chronological ordering preserves the temporal sequence of comments but may include many redundant or low-value responses before reaching informative evidence. Deep-shallow ordering emphasizes deeper conversational chains, which can capture extended discussions but may still prioritize comments that are structurally deep rather than evidentially useful. As shown in Fig. 10, EDM achieves stronger performance by selecting compact evidence comments according to both source relevance and lexical informativeness. This result suggests that the benefit of EDM does not come from merely reordering comments, but from filtering the comment stream into more decision-relevant evidence for LLM reasoning.

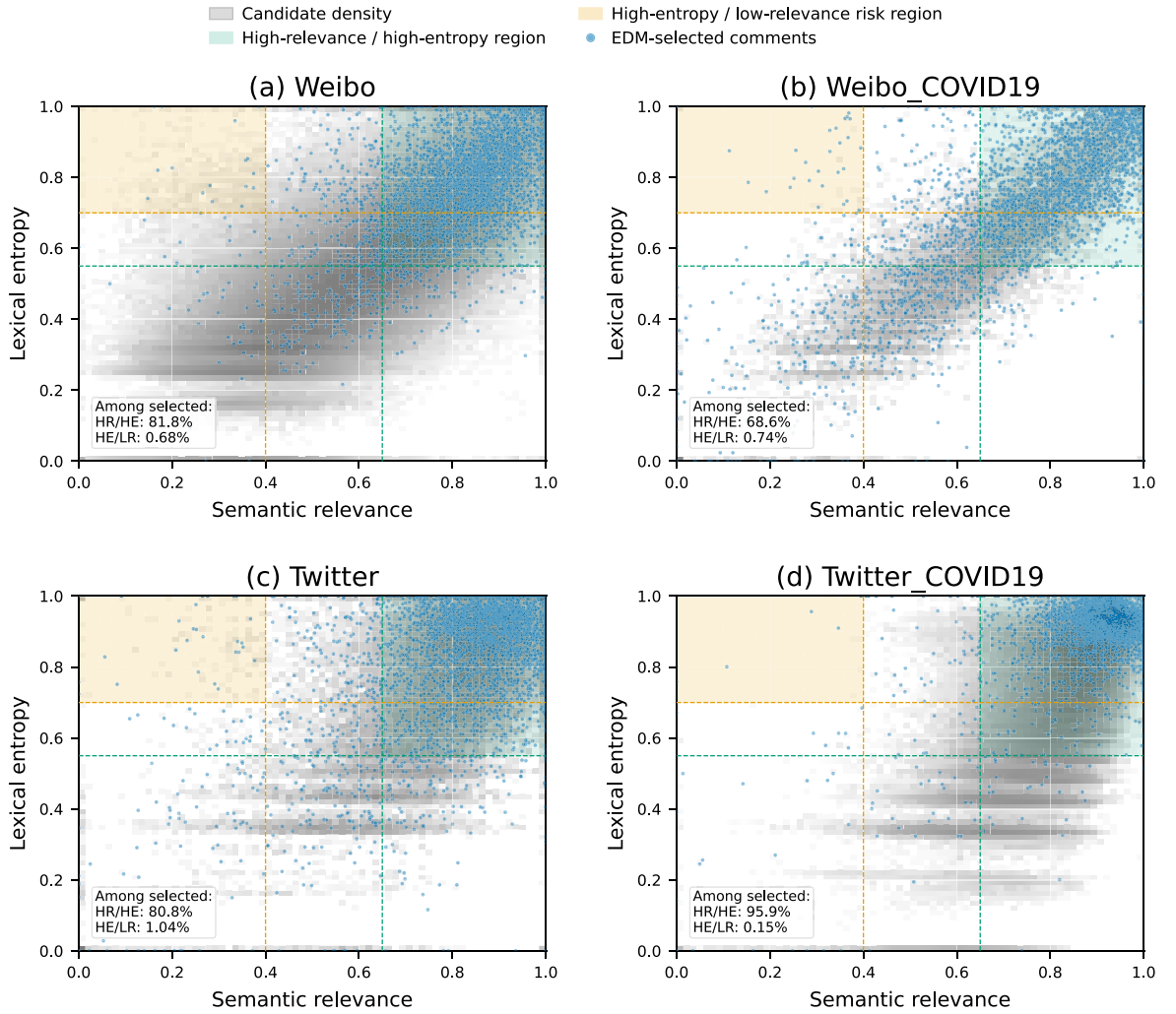


Fig. 9. EDM selection in the semantic relevance–lexical entropy space. Panel annotations report the percentages of EDM-selected comments in the high-relevance/high-entropy (HR/HE) and high-entropy/low-relevance (HE/LR) regions, showing that selected comments concentrate in HR/HE regions while rarely falling into HE/LR risk regions.

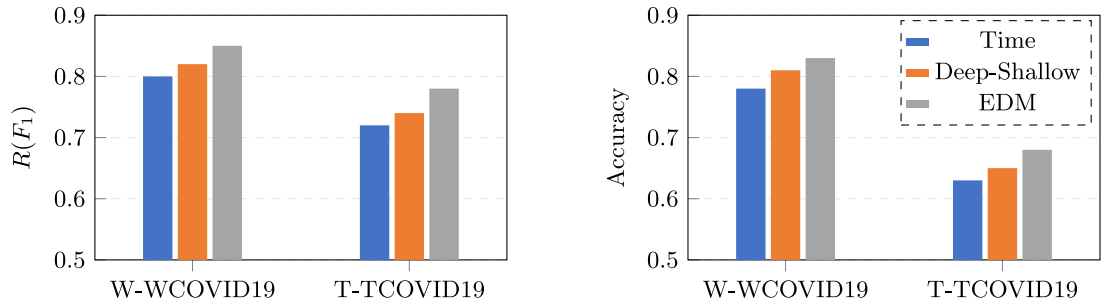


Fig. 10. An experimental study of comment organization strategies.

6.3. Robustness and sensitivity analysis

6.3.1. Sensitivity and thread-scale robustness of key hyperparameters

This section examines the sensitivity of RumorGAL to three key hyperparameters: the number of structural anchors k in the SA module, the number of retained evidence comments q in the EDM module, and the trade-off factor α in the evidence density score. Performance is evaluated using $R(F_1)$ on both transfer settings.

Table 7
EDM decisions for emoji-heavy, repetitive, and promotion-like comments.

Type	Representative comment with highlighted cue	Rel.	Ent.	EDM
Emoji-heavy comments				
Source post: "Parliament Hill attack: Ottawa residents captured images of chaos amidst lockdown".				
Emoji and character-stuffing noise	"[emoji/number/character sequence] ev3 2 1 ...qqqq ...kkkm"	0.000	0.817	✗ Drop
Source-relevant event context	"Comedy amongst the chaos: fifteen members walking around; two officers crouched behind a PC".	1.000	0.952	✓ Keep
Source post: "President made a commitment to donate his salary while in office. Honoring that promise ..."				
Emoji-alias stuffing	"grinning_face grinning_face_with_big_eyes ...face_with_tears_of_joy ...somebody tell him to get his golf clubs ..."	0.270	0.910	✗ Drop
Source-relevant alias comment	"You've spent more of our tax dollars on drinks than that. 100 million to golf straight in his pocket. clown_face"	0.911	0.985	✓ Keep
Repetitive comments				
Source post: "Fox News claims it never called COVID-19 a hoax. Let's check the receipts".				
Spaced-letter repetition	"Fox, watch your own tapes. I will say it s l o w l y: w a t c h y o u r o w n t a p e s".	0.068	0.971	✗ Drop
Claim-related evidence	"Trump was first to call coronavirus a hoax. At a rally, I think; remember, he said: this is just ..."	0.973	0.950	✓ Keep
Source post: "Islamist militants take hostages in Sydney cafe, display black jihadist flag".				
Repeated-character distortion	"ITS NOTTTTTT... a jihadist flaqqqqqq... imbielssss... priestttttt..."	0.136	0.845	✗ Drop
Source-relevant correction	"It is NOT a Jihadist flag. It is an Islamic declaration about Allah and Mohammed that has been hijacked by extremists".	1.000	0.931	✓ Keep
Promotion-like comments				
Source post: "As reports come in from Sydney, remember that news outlets will get things wrong".				
Product-ad noise	"Share views, friends; Daikin ceiling-cassette air conditioner".	0.000	0.797	✗ Drop
Source-relevant repost	"As reports come in from Sydney, remember that news outlets will get things wrong".	1.000	0.797	✓ Keep
Source post: "Here is why steps are being taken to ban solitary confinement for low-level infractions".				
Channel-promotion cue	"Hey check my channel; if you like, please subscribe for more shocking videos in future. Thanks".	0.558	0.857	✗ Drop
Policy-relevant comment	"I second the idea to eradicate solitary confinement; it is associated with a lot of suicide attempts".	1.000	0.949	✓ Keep

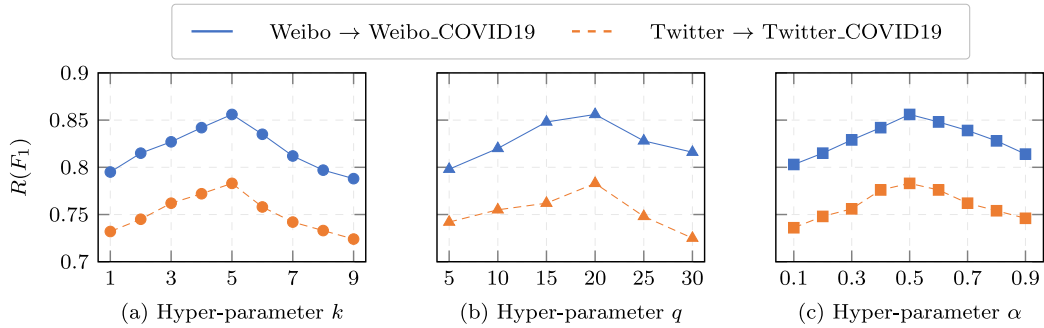


Fig. 11. Sensitivity analysis of RumorGAL with respect to key hyperparameters.

As shown in Fig. 11(a), performance peaks when k is around 5, suggesting that a moderate number of structural anchors is sufficient to capture useful structural cues. Smaller values may miss relevant local interactions, whereas larger values may introduce noisier or less relevant structural information. Fig. 11(b) shows that the best performance is achieved when q is around 20, indicating that the LLM benefits from a compact yet informative evidence set. Too few comments may provide insufficient context, while too many comments may dilute key evidence with redundant or irrelevant discussion content. As shown in Fig. 11(c), the best trade-off is obtained when α is close to 0.5, suggesting that semantic relevance and lexical informativeness should be balanced in evidence selection. Lower values place greater emphasis on lexical informativeness and may introduce less relevant comments, whereas higher values overemphasize semantic similarity and may reduce evidence diversity.

To further examine whether the fixed setting of $k = 5$ and $q = 20$ is affected by thread scale, we evaluate RumorGAL on very small threads with $n < 10$ and very large threads with $n > 500$. As shown in Table 8, RumorGAL maintains stable performance across both scale groups and transfer settings. For small threads, the selection process remains score-driven, with SA ranking available nodes by their cosine similarity to the source-post node and EDM ranking available comments by evidence density. For large threads, the fixed budget serves as a compact structural and evidence filter, selecting key cues for the LLM while reducing redundant context and input-length pressure.

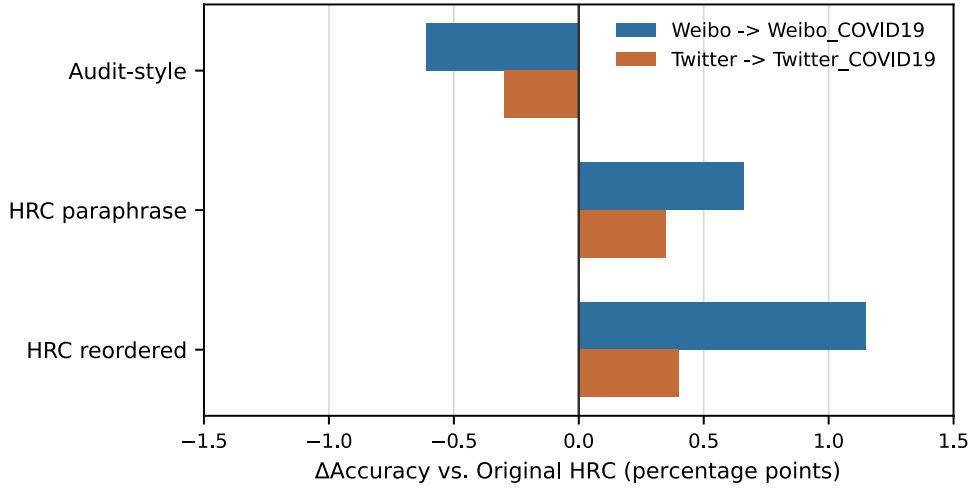
Table 8Thread-scale robustness under the fixed budget setting of $k = 5$ and $q = 20$.

Transfer setting	Thread group	Events	Acc.	R.(F ₁)
Weibo → Weibo_COVID19	$n < 10$	121	0.8678	0.9070
	$n > 500$	15	0.8667	0.8826
Twitter → Twitter_COVID19	$n < 10$	16	0.7625	0.7852
	$n > 500$	122	0.7759	0.8071

Table 9

Instruction variants used in the prompt sensitivity analysis.

Prompt variant	Instruction difference
Original HRC	Uses the original hierarchical rumor-checking instruction, including source-claim examination and comment-evidence evaluation.
Audit-style instruction	Uses a flat checklist of rumor-risk cues, without separating source-claim verification from comment-evidence integration as in HRC.
HRC paraphrase	Keeps the same HRC reasoning logic but rewrites the instruction with semantically equivalent wording.
HRC reordered clues	Keeps the HRC reasoning factors but changes the evidence order by examining comment evidence before source-post verification.

**Fig. 12.** Prompt sensitivity measured by the change in accuracy relative to the Original HRC prompt ($\Delta\text{Acc.}$, percentage points). The vertical line at zero denotes the Original HRC baseline.

6.3.2. Prompt sensitivity analysis

To examine whether HRC depends on a particular prompt wording, we conduct a prompt sensitivity analysis by modifying the instruction part while keeping the model checkpoint, input evidence, structural prefixes, output format, and decoding setting unchanged. Table 9 summarizes the tested variants, including the original HRC prompt, an audit-style checklist, an HRC paraphrase, and an HRC variant with reordered evidence clues. Fig. 12 reports accuracy changes relative to the original HRC prompt, where the vertical zero line denotes the original HRC performance. The HRC-preserving variants, including the paraphrased and reordered prompts, remain close to or slightly above the zero line on both transfer settings, indicating that HRC shows limited sensitivity to surface wording and evidence order. In contrast, the audit-style checklist falls below the original HRC, suggesting that a flat checklist is less effective than the hierarchical organization of source-claim verification and comment-evidence integration.

6.3.3. Effect of LLM backbone capacity

To examine whether structural anchoring remains useful when the base LLM becomes stronger, we replace DeepSeek-R1-7B with Qwen2-72B while keeping SA, EDM, HRC, and the training protocol unchanged. As shown in Table 10, Qwen2-72B improves the absolute performance of both the w/o SA variant and RumorGAL. More importantly, RumorGAL still outperforms the corresponding

Table 10

Effect of LLM backbone capacity under cross-domain rumor detection.

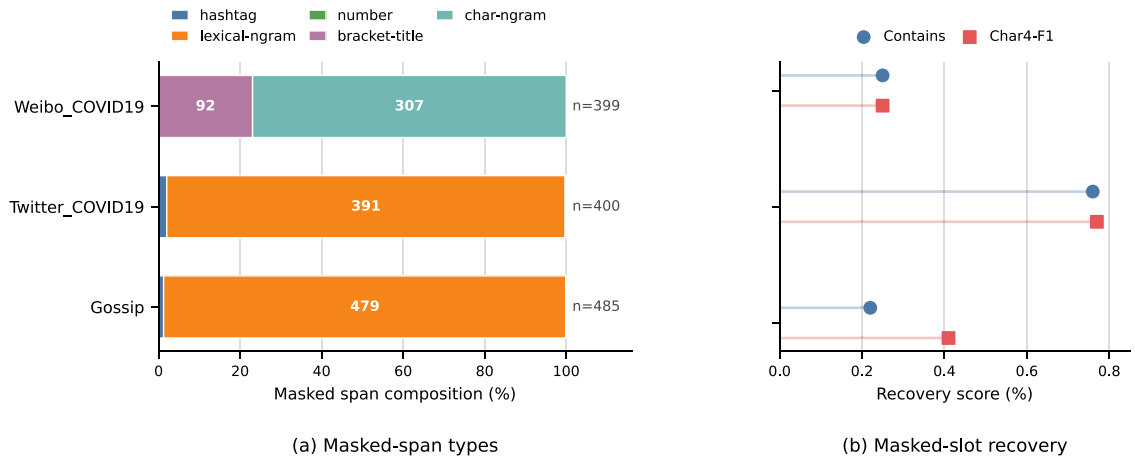
Dataset	Backbone	Setting	Acc.	R(F ₁)	Δ Acc.	Δ R(F ₁)
Weibo→Weibo_COVID19	DeepSeek-R1-7B	w/o SA	0.8118	0.8204	–	–
	DeepSeek-R1-7B	RumorGAL	0.8328	0.8568	+0.0210	+0.0364
	Qwen2-72B	w/o SA	0.8345	0.8692	–	–
	Qwen2-72B	RumorGAL	0.8540	0.8833	+0.0195	+0.0141
Twitter→Twitter_COVID19	DeepSeek-R1-7B	w/o SA	0.6731	0.7663	–	–
	DeepSeek-R1-7B	RumorGAL	0.6822	0.7835	+0.0091	+0.0172
	Qwen2-72B	w/o SA	0.6892	0.7845	–	–
	Qwen2-72B	RumorGAL	0.6975	0.7924	+0.0083	+0.0079

Table 11

Suffix-completion data-contamination probe using DeepSeek-R1-7B. The model receives the raw prefix of each source post and generates the continuation.

Dataset	Prefix	Completion		Leakage indicator		Random baseline		Adjusted	
		C4	C4@50	Exact	NC@50	R-C4	R@50	Adj-C4	Adj@50
Weibo_COVID19	30%	0.0116	0.0152	0.0000	0.0025	0.0003	0.0002	0.0113	0.0150
	50%	0.0100	0.0119	0.0000	0.0000	0.0004	0.0006	0.0096	0.0113
	70%	0.0096	0.0151	0.0000	0.0050	0.0001	0.0001	0.0095	0.0150
Twitter_COVID19	30%	0.0191	0.0162	0.0000	0.0000	0.0124	0.0046	0.0067	0.0116
	50%	0.0202	0.0235	0.0000	0.0000	0.0103	0.0068	0.0099	0.0167
	70%	0.0174	0.0283	0.0000	0.0025	0.0076	0.0064	0.0098	0.0219
Gossip	30%	0.0115	0.0094	0.0000	0.0000	0.0099	0.0045	0.0016	0.0050
	50%	0.0096	0.0114	0.0000	0.0000	0.0089	0.0053	0.0007	0.0061
	70%	0.0091	0.0122	0.0000	0.0000	0.0056	0.0038	0.0035	0.0085

Notes. C4 denotes the character-level 4-gram F₁ between the generated continuation and the gold suffix. C4@50 computes the same score on the first 50 generated characters. Exact denotes the exact-match rate between the generated continuation and the gold suffix. NC@50 denotes the near-copy rate based on C4@50. R-C4 and R@50 denote the corresponding random-suffix baselines for C4 and C4@50, respectively. Adj-C4 and Adj@50 denote the adjusted scores obtained by subtracting the corresponding random-suffix baselines from C4 and C4@50.

**Fig. 13.** Masked-slot data-contamination probe. Left: composition of masked span types across datasets. Right: recovery scores for masked spans.

w/o SA variant under the 72B backbone on both transfer settings. This result indicates that the benefit of structural anchoring is not limited to smaller LLMs, and that propagation-aware structural cues remain useful as auxiliary information in the large-backbone setting. These results support our “LLM-as-primary, GNN-as-auxiliary” design across different model capacities.

6.4. Data contamination bias and mitigation

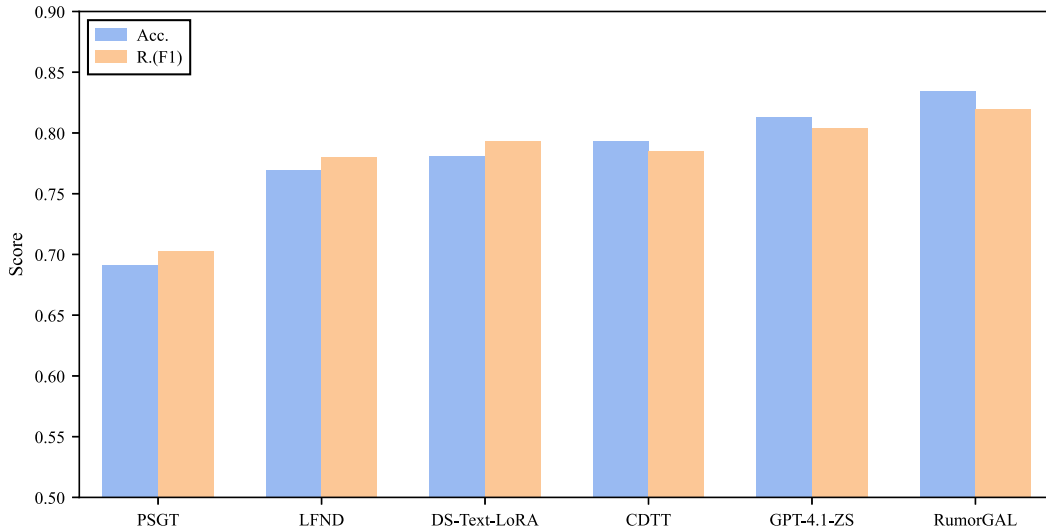
Contamination-risk probing. Public benchmark contamination is a known threat to LLM-based evaluation, especially for rumor detection benchmarks constructed from public social-media posts that may have been exposed during model development (Xu et al., 2024). Such exposure may make performance on public target-domain datasets partially reflect benchmark memorization rather than true cross-domain generalization. Because direct instance-level verification is infeasible without access to LLM training corpora, we follow prior black-box contamination studies (Golchin & Surdeanu, 2024) and adopt suffix completion and masked-slot recovery as two indirect behavioral diagnostic probes.

Table 12

Effect of excluding flagged near-copy instances.

Target dataset	Flagged (%)	$\Delta\text{Acc.}$	$\Delta\text{R-F}_1$
Weibo_COVID19	0.75	$+0.0016 \pm 0.0022$	$+0.0035 \pm 0.0026$
Twitter_COVID19	0.25	$+0.0006 \pm 0.0008$	$+0.0001 \pm 0.0004$

Notes. Flagged denotes the percentage of target-domain instances removed when constructing the clean subset. An instance is flagged as a potential near-copy if it satisfies $\text{SeqRatio}_p \geq 0.8$, $\text{C4}_p \geq 0.5$, or $\text{C4@50}_p \geq 0.5$ under any prefix p . $\Delta\text{Acc.}$ and $\Delta\text{R-F}_1$ are computed as Full minus Clean. Values close to zero indicate that removing flagged instances has negligible impact on the evaluation results.

**Fig. 14.** Emerging-event comparison on Weibo-2026H1.

We apply both probes to the backbone LLM using greedy decoding. The suffix-completion probe evaluates whether the model can reproduce the unseen continuation of a source post when given its raw prefix. Specifically, we use 30%, 50%, and 70% prefixes and compare the generated continuation with the gold suffix using character-level 4-gram F1, exact match, near-copy rate, and random-baseline-adjusted scores. If target instances were memorized verbatim, longer prefixes would be expected to trigger high suffix recovery or near-copy behavior. However, Table 11 shows that exact match is zero across all datasets and prefix ratios. The near-copy rate remains at most 0.0050, and the largest Adj@50 score is 0.0219, indicating no substantial suffix-level memorization signal.

We further conduct a complementary masked-slot recovery probe to examine instance-specific memorization. This probe masks rare or identifying spans, including hashtags, numbers, bracketed titles, lexical n-grams, and character n-grams, and measures whether the model can recover the missing span. As shown in Fig. 13, the masked spans cover different instance-specific cues across datasets, while recovery scores remain consistently low. Character-level 4-gram F1 is also below 1% across datasets. Overall, these diagnostics provide no evidence of substantial verbatim or slot-level memorization in the evaluated public target instances.

Clean and temporal mitigation. To mitigate the potential influence of contaminated public benchmarks, we first construct clean target-domain subsets by removing all instances flagged as potential near-copy cases according to the contamination probes, excluding Gossip where no such cases are detected. RumorGAL is then re-evaluated on these cleaned subsets using the same trained models and evaluation protocol. As shown in Table 12, the flagged ratio is below 1% for all target datasets with detected near-copy cases, and the performance differences between the full and clean evaluations are consistently negligible. This suggests that the main benchmark conclusions are not materially driven by the small number of potentially memorized near-copy instances.

We further constructed *Weibo-2026H1*, a temporally held-out dataset, to evaluate the generalization of RumorGAL to emerging events. As shown in Fig. 14, RumorGAL maintains stable performance on this temporally recent event set. To construct *Weibo-2026H1*, we selected public events occurring in the first half of 2026 and formulated event-specific queries based on their core entities, locations, and claims. Candidate posts were retrieved through Weibo Search.³ The collected events cover U.S.–Iran tensions, the White House Correspondents’ Dinner shooting, incidents involving Zhang Xue motorcycles, and other emerging public events. After filtering and deduplication, the dataset contains 573 public discussion threads, including 326 rumors and 247 non-rumors. For each candidate instance, we used LLM-assisted web search to retrieve potentially relevant evidence from official statements,

³ <https://s.weibo.com/>

Table 13

Training and inference cost comparison.

Dataset	Backbone/System	Peak memory (GiB/GPU)	Train GPUs	Deploy GPUs	Train time (min)	Latency (ms)
Weibo→Weibo_COVID19	DeepSeek-R1-7B LoRA	≤20	2	1	115	2100
	Qwen2-72B 4-bit LoRA	≤22	6	3	176	3000
	DeepSeek-R1-7B Full	OOM	2	–	–	–
	deepseek-reasoner API	–	–	API	–	2500
Twitter→Twitter_COVID19	DeepSeek-R1-7B LoRA	≤19	2	1	65	1700
	Qwen2-72B 4-bit LoRA	≤22	6	3	154	2600
	DeepSeek-R1-7B Full	OOM	2	–	–	–
	deepseek-reasoner API	–	–	API	–	2300

reputable news reports, and established fact-checking sources. Human annotators subsequently verified the retrieved evidence and conducted additional manual searches when necessary. Instances with insufficient evidence or ambiguous claims were excluded. Each retained instance was independently labeled by three annotators, with the final label determined by majority voting. A description of *Weibo-2026H1* is available in the project repository.⁴ Given the privacy considerations associated with social-media data, the complete dataset will be made available through controlled access for research purposes, subject to the submission of basic applicant information and acceptance of the corresponding data-use agreement.

6.5. Efficiency and case study

6.5.1. Efficiency and cost analysis

Table 13 reports the training and inference costs of RumorGAL under different backbone capacities and deployment settings. We compare the default DeepSeek-R1-7B backbone with a larger Qwen2-72B backbone using 4-bit LoRA fine-tuning. We also include DeepSeek-R1-7B full fine-tuning under the same local training setting and API-based inference using `deepseek-reasoner`. The results show that LoRA enables efficient adaptation with moderate computational cost. With the 7B backbone, RumorGAL can be fine-tuned within 20/19 GiB peak memory per GPU on the two transfer settings and deployed locally on a single GPU. Scaling to Qwen2-72B increases training time and inference latency, but remains feasible with 4-bit LoRA and multi-GPU deployment. In contrast, full fine-tuning leads to out-of-memory failures under the same two-GPU training setting, highlighting the importance of parameter-efficient adaptation. Compared with API-based inference, the local 7B deployment achieves lower latency under our measured setting. Overall, RumorGAL achieves a practical balance between model capacity and computational cost.

6.5.2. Case study

Table 14 presents a post-hoc qualitative analysis of two representative cases. In the rumor case, the model captures suspicious cues in the source post, including absolutist claims and pseudo-scientific expressions, while the comments selected by EDM provide additional evidence such as refutations and personal experiences. Under the HRC prompting framework, these signals are organized into a coherent reasoning process that supports the rumor prediction. In the Non-Rumor case, the model identifies scientifically grounded health advice in the source post and further verifies it through supportive comments selected from the discussion. The resulting reasoning process suggests that the prediction is jointly supported by the source content and the selected comments.

7. Discussion and implications

The experimental results consistently support the central design of RumorGAL. The overall comparison shows that combining propagation-aware structural cues with LLM reasoning improves generalization beyond text-only and propagation-based models, while the ablation and mechanism analyses confirm that structural anchoring, evidence selection, and hierarchical reasoning provide complementary contributions. The robustness and backbone analyses further indicate that these gains are not tied to a specific prompt wording, thread scale, or LLM capacity. Meanwhile, the weaker performance on Terrorist→Gossip reveals an important boundary: the transferability of propagation patterns can be constrained when source-domain diversity is limited. Taken together, these findings suggest that robust domain-generalized rumor detection benefits from grounding LLM reasoning in compact structural cues and informative evidence rather than relying on either textual reasoning or propagation modeling alone.

Theoretical Implications. This work provides a new perspective on domain-generalized rumor detection by showing that propagation structure can serve not only as a predictive representation for graph-based classifiers, but also as an auxiliary reasoning prior for LLMs. This finding suggests a complementary division of labor between structural modeling and semantic reasoning: the graph encoder captures transferable propagation cues, while the LLM performs evidence-grounded interpretation and decision-making. More broadly, the results indicate that domain generalization can be improved not only through representation alignment or invariant feature learning, but also by grounding LLM reasoning in structural signals that are less dependent on event-specific textual patterns.

⁴ <https://github.com/RumorResearch/Weibo2026H1>

Table 14

Post-hoc qualitative analysis of representative rumor and non-rumor cases.

Source Post: “Corona virus pro tip: if you mix Emergen-C with vodka you literally cannot get sick, it’s actually science”.
Key Clues: – “I’ve been doing this for years and I’m always sick so”. – “Not true unfortunately. You need 60% alcohol mixture for true hand sanitize”. – “Us all weekend, can attest: not a successful course of action”.
Label (Prediction/Ground Truth): Rumor/Rumor
Post-hoc Qualitative Analysis: Overall Assessment: By integrating global and local cues, the model accurately identifies the post as a rumor. Source Post Analysis: The post makes an absolute claim (“literally cannot get sick”) and attempts to legitimize it with a scientific veneer, which contradicts established medical knowledge about virus transmission and prevention. Comment Comparison: Comments consistently challenge the post’s claim, offering both scientific corrections (e.g., required alcohol concentration for sanitization) and personal counterexamples, reflecting widespread user skepticism.
Source Post: “COVID-19 is most contagious within 5 days of symptoms onset. Do not let your guard down, and avoid a lucky mindset. The virus is too cunning. Wear a mask, wash hands frequently, exercise appropriately, adjust your daily schedule, avoid consuming imported frozen meat and seafood, and avoid crowded places. In short, take proper precautions and don’t let your guard down, don’t relax”.
Key Clues: – “Take precautions when going out, protect yourself”. – “Everyone should be aware”. – “Don’t let your guard down”.
Label (Prediction/Ground Truth): Non-Rumor/Non-Rumor
Post-hoc Qualitative Analysis: Overall Assessment: By synthesizing global and local signals, the model correctly classifies the post as a Non-Rumor. Source Post Analysis: The post shares scientifically grounded preventive measures—mask-wearing, hand hygiene, and avoiding crowds—consistent with public health guidelines and devoid of exaggeration or misinformation. Comment Comparison: Comments largely affirm the post’s message, reinforcing its validity through supportive statements and additional precautionary suggestions, with no contradictory or skeptical responses observed.

Practical Implications. RumorGAL is particularly applicable to emerging-event rumor detection under a strict domain-generalization setting, where target-domain samples are unavailable for training or adaptation. By learning exclusively from source-domain data, the framework avoids target-domain alignment while maintaining robust cross-domain generalization. Moreover, graph-anchored structural prefixes compactly encode propagation cues, and evidence density-guided comment selection retains informative evidence under a limited context budget; LoRA-based parameter-efficient fine-tuning further supports resource-efficient local deployment.

Limitations and future work This work evaluates RumorGAL under domain generalization settings and demonstrates its effectiveness across standard benchmarks and emerging events. However, RumorGAL currently relies on fixed evidence budgets and label-free evidence signals based on semantic relevance and lexical informativeness, leaving room for adaptive evidence allocation and richer evidence modeling in noisy social-media contexts. Moreover, the interaction between propagation structures and LLM reasoning could be further strengthened beyond the current lightweight graph-derived prefix mechanism. Our boundary analysis further shows that limited source diversity can hinder the transfer of propagation patterns to unseen target events, highlighting the need for more diverse training scenarios and adaptive strategies in evolving misinformation environments.

8. Conclusion

Rumors often emerge during real-world breaking events, where rapidly evolving information and unseen scenarios pose significant challenges for reliable detection. This paper studies rumor detection in domain generalization scenarios and proposes RumorGAL, a graph-anchored LLM reasoning framework that jointly leverages propagation structure and semantic evidence. Experimental results on multiple benchmarks demonstrate the effectiveness and robustness of RumorGAL under domain generalization scenarios.

CRedit authorship contribution statement

Lina Han: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Yuhan Qiao:** Writing – review & editing, Visualization. **Feiyang Jia:** Writing – review & editing. **Yan Zhao:** Writing – review & editing. **Caiyan Jia:** Writing – review & editing, Resources, Project administration, Funding acquisition, Conceptualization.

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Data availability

The authors do not have permission to share data.

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